

* if the charges placed in a medium having relative permittivity $\epsilon_r \Rightarrow K$

$$F = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2} \quad \textcircled{i} \quad \text{in space}$$

$$F' = \frac{1}{4\pi\epsilon_0 \epsilon_r} \frac{q_1 q_2}{r^2} \quad \textcircled{ii} \quad \text{in medium}$$

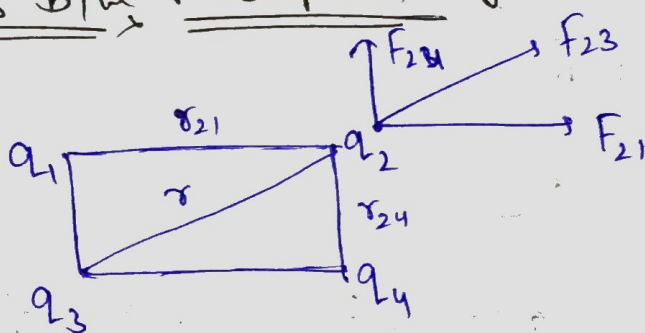
$\epsilon_m = \epsilon_0 \epsilon_r$

$$F' = \frac{F}{\epsilon_r} \quad \boxed{\epsilon_r \Rightarrow K}$$

$F' = \frac{F}{K}$

Decreases by K times

Forces b/w multiple charges



force on any charge due to a number of other charges is the vector sum of all the forces on that charge due to other charges.
 $F_{23} \Rightarrow$ Force on 2 due to 3 etc.

* The individual forces are unaffected due to the presence of other charges.
 This is termed as the "Principle of Superposition"